Nishal Stanislaus Silva, *PhD*

Machine Learning Engineer | Production ML Systems, MLOps on AWS, Time-Series | Applied ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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18th October 2026

Re: Senior Data Scientist – AutoTrader.ca

Dear Hiring Manager,

I am applying for the Senior Data Scientist role at AutoTrader.ca. I am a Machine Learning Engineer with a PhD (University of Trento, 2025) and 10+ years across industry and academia, and I own ML systems end-to-end in production: problem framing, design, deployment, monitoring and iteration. At MAS Holdings I led computer vision and IoT systems for manufacturing that cut inspection time by 99.5% and raised operational efficiency by up to 300%, working with product, operations and engineering stakeholders to turn business needs into measurable results.

Three parts of your posting match my work directly. For production ML, my published system runs at 14 ms and F1 0.76 on a Raspberry Pi 4, 74x faster than the DTW baseline (Journal of the Audio Engineering Society, 2026), and I owned its lifecycle from data acquisition to deployment and monitoring. For AWS and MLOps, I have built Airflow, Docker and ETL pipelines on AWS (S3, ECS, MWAA, RDS) with Snowflake, and at MAS I introduced CI/CD and code review practice and mentored engineers. For methodology, I design benchmark suites with baselines and ablations to guide architecture decisions under compute constraints, and I released four open datasets on Zenodo.

I should be direct about two gaps. My GenAI experience is applied and tool-level (RAG, prompt engineering, and diffusion and VAE based synthetic data work at McGill), not LLM research, and I have not yet operated ML systems across several countries or markets. I have shipped systems across distributed factory sites and I expect to learn the regional differences quickly. What draws me to AutoTrader is a marketplace where ML sits at the core of the product and the work is judged by outcomes, not models. Within the first months I would aim to contribute to production reliability and monitoring of your ML systems.

I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, and available immediately. I would welcome the opportunity to discuss how my experience can support your Data Science team at AutoTrader.ca. My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD